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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CPE 324 — Past Questions & Materials

by Renance

Course: CPE 324

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — CPE 324

Level 300 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.

COLLEGE OF ENGINEERING AND TECHNOLOGY

th DEPARTMENT OF COMPUTER ENGINEERING

4 XAM: 2nd Semester

eu. COURSE TITLE: Computer Programming with C/C++

SESSION: 2024/2025 Os COURSE CODE: CPE 324

DATE: 02/10/2025 ES TIME ALLOWED: 3 Hours

INSTRUCTIONS: Answer any five (5) questions. Show all necessary workings and descriptions. No markings, notes, or inscriptions on the question paper is allowed.

- Qta. The following C++ program contains errors. Provide the corrected program and underline the corrected portions. 6Mks

```
#include <stream>

int main

{cout << "If you see this message",
cout >> " your corrections are in order,"
cout << " endl;"

cout << 'Use an umbrella to stay dry'
<< " because it might rain!" << endl.

return 0;
```

Qib. Write a C++ program that displays the memory space required by the data types float, short, double,

and signed char on screen. 5Mks

Qic. What does the statement “C is backward compatible with C++ mean? 3Mks

~ Q2a. Write a C++ program to receive a person’s name, displays the name and length of the name. 5Mks

Q2b. Write a C++ program that receives a decimal number and outputs the octal, decimal, and hexadecimal values. 5Mks

Q2c. Using knowledge of operator precedence, what would the solution of the expression

Int num(5); cout << num++ — 6/2: be? 4Mks

*Q3a. Given the C++ program below state the values of the variables for each line of cout code. 5Mks

```
#include <iostream>

using namespace std;

int main()

{int i(3), j(7);

cout << i++ << endl;
```

```
cout << i << endl;
cout << j-- << endl;
cout << --j << endl;
return 0;}
```

Q3b. Specify true or false for the following relational operator expressions. (i) $5 \geq 6$ (ii) $1.7 < 1.8$
(ii) $4 + 2 == 5$ (iv) $2 * 4 != 7$. 5Mks

Q3c: List the four unary operators in C/C++. 4Mks

Q4a. Distinguish C++ implicit and explicit data type conversion in function calls using examples. 6Mks

Q4b. Write a C++ program that reads several lines of text and outputs them in reverse order. 5Mks

Q4c. Describe compound assignment operators in C++. 3Mks

Q5a: Distinguish between string concatenation with "+" and "+=" in C++. 5Mks

Q5b. Write a C++ program that calls a function to calculate and display the area of a rectangle. 6Mks

Q5c. Describe function overloading in C++ integrating example lines of code. 3Mks

Q6a. Write a C++ program that calls a function with pointer input arguments to swap two floating point

values. 5Mks ; ;

Q6b. Write a C++ program that defines one integer and one pointer variables respectively then prints their values and memory addresses. 5Mks .

Q6c. (i) Why is it useful to declare C/C++ functions in a header file? (ii) What is the purpose of the void return

type in functions? 4Mks

Q7a. Describe four characteristics of classes (also known as objects) in C++. 5Mks

Q7b. Define a simple C++ class for bank accounts distinguishing its private and public members 4 header file. Add a text description of the program flow below the code. 6Mks

Q7c. Using examples describe two ways the members of a class can be accessed. 3Mks